| Sr. No. | Title of Experiment | Page NO | A2 batch | Sign |
|---|---|---|---|---|
| | **BI** | | | |
| 1 | **DNA Sequence Analysis:** Analyze a given DNA sequence and perform basic sequence manipulation, including finding motifs, calculating GC content, and identifying coding regions | | Jul 09,2026 | |
| 2 | **RNA-Seq Data Analysis:** Analyze a provided RNA-Seq dataset and perform differential gene expression analysis. | | Jul 15,2026 | |
| 3 | **Protein Structure Prediction:** Predict the 3D structure of a given protein sequence using homology modeling or threading techniques. | | Jul 21,2026 | |
| 4 | **Molecular Docking and Virtual Screening:** Perform molecular docking simulations to predict the binding affinity between a protein target and a small molecule ligand. Additionally, conduct virtual screening to identify potential drug candidates. | | Jul 27,2026 | |
| 5 | **Machine Learning for Genomic Data**: Apply machine learning algorithms, such as random forests or support vector machines, to classify genomic data based on specific features or markers. | | Aug 03,2026 | |
| 6 | **Agricultural Genomics and Crop Improvement**: Analyze genomic data from crops to identify genetic markers associated with desirable traits, such as disease resistance or yield. | | Aug 10,2026 | |
| | **IR** | | | |
| 7 | Write a program for pre-processing of a text document such as stop word removal, stemming. | | Aug 14,2026 | |
| 8 | Implement a program for retrieval of documents using inverted files. | | Aug 21,2026 | |
| 9 | Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set (You can use Java/Python ML library classes/API. | | Aug 26,2026 | |
| 10 | Implement Agglomerative hierarchical clustering algorithm using appropriate dataset. | | Sep 04,2026 | |
| 11 | Implement Page Rank Algorithm. (Use python or beautiful soup for implementation). | | Sep 11,2026 | |
| | **Mini Project** | | | |
| 12 | Mini Project: | | Sep 24,2026 | |